10.11.1 IEC Data File Contents

The impedances and admittances of electrical equipment in the working case are modified according to the correction factors defined in IEC 60909-0 (2016, Section 6 or 2001, Section 3).

Generators, equivalent generators, asynchronous motors, wind farms and photovoltaic (PV) farms are represented as sources in the working case. To calculate their impedance correction factors or correct contribution to fault additional data is required. This additional data is provided in IEC data file (.iec).

Additional data is needed in the following cases.

- If a generator model includes a GSU transformer. It is recommended to represent the GSU transformer explicitly as a separate power component so as to be able to correctly modify the generator and transformer impedances as per the IEC 60909 standard.
- If a generator in the working case is an equivalent generator representation.
- If a generator is a used in the modeling of a synchronous motor.
- If a generator is set with QMIN=QMAX, then it is treated as an asynchronous motor. If this is the case, then additional data is not necessary.
- If a transformer winding MVA specified in system MVA base and not nameplate winding MVA.
- If an induction machine base power (MBASE) is specified as mechanical power output (BCODE=1), then power factor and efficiency are needed to calculate base MVA. If an induction machine scheduled active power (PSET) is set as real electrical power drawn by the machine, efficiency is needed to calculate mechanical power output (MW) of the machine.
- If a generator in the working case is a Wind Farm or PV Farm.

There are six groups of records, with each group containing a particular type of data required. Each record is terminated by specifying a bus number of 0.

GSU, Equivalent Generator and Induction Machine Data

(Induction Motors are specified as part of Generator data category.)

Each data record has the following format:

I, ID, MCTYPE, UrG, PG, PFactor, PolePair, GSUType, Ix, Jx, Kx, Ckt, PT

where:

I	Machine bus number
ID	Machine ID

MCTYPE	Machine type MCTYPE =1, for Generator MCTYPE =2, for Equivalent generator MCTYPE =3, for Induction machines specified as part of generator data category MCTYPE =4, for Wind power station asynchronous generator MCTYPE =5, for Wind power station double fed asynchronous generator MCTYPE =6, for Wind or Photovoltaic power station generator with full size converter
UrG	Rated terminal voltage, line-to-line in kV r.m.s. (this need not be the rated bus voltage) = 0.0 by default to treat rated voltage same as bus voltage
PG	Range of generator voltage regulation in %, e.g., if PG is ±5%, enter PG=5. PG = (UG -UrG)/UrG, where UG is the scheduled generator terminal voltage = 0 default
PFactor	Generator rated power factor (used only if MCTYPE=1). This is used in impedance correction factor calculations. = 1.0 default
PolePair	Number of pole pairs if machine is induction machine (used only if the machine is modeled as induction machine) Example: If the induction machine has a six pole construction then Polepair=3
GSUType	Generator step-up-transformer type GSUType =0, no GSU, GSU transformer modeled explicitly. GSUType =1, GSU with OLTC GSUType =2, GSU without OLTC
Ix	GSU transformer I bus number = 0 by default to specify no GSU transformer
Jx	GSU transformer J bus number = 0 by default to specify no GSU transformer
Kx	GSU transformer K bus number = 0 by default to specify no GSU transformer
Ckt	GSU transformer circuit identifier = ‘’ by default to specify no GSU transformer
PT	Tap range of GSU off-load tap-changer transformer in % of transformer winding rated voltage, e.g., if PT is ±5%, enter PT=5 = 0.0 by default to ignore off-load tap range

Transformer Nameplate Winding MVA Data

Each data record has the following format:

I, J, K, Ckt, Sbase 1-2, Sbase 2-3, Sbase 3-1

where:

I	Winding 1 bus number
J	Winding 2 bus number
K	Winding 3 bus number (=0 for two-winding transformer)
Ckt	Transformer circuit identifier
Sbase 1-2	Winding 1 to winding 2 Nameplate MVA
Sbase 2-3	Winding 2 to winding 3 Nameplate MVA (not required for two-winding transformer)
Sbase 3-1	Winding 3 to winding 1 Nameplate MVA (not required for two-winding transformer)

Induction Machine Data

(Induction Motors are specified as part of Induction machine data category.)

Each data record has the following format:

I, ID, PolePair, PFactor, Efficiency

where:

I	Machine bus number
ID	Machine ID
PolePair	Number of pole pairs Example: If the induction machine has a six pole construction then Polepair=3
PFactor	Induction Machine rated power factor = 1.0 default
Efficiency	Induction Machine percent efficiency. =100 by default Example: Efficiency=96.5 for 96.5% efficiency.

Wind Power Station Asynchronous Generator Data

(These equipment are specified as part of Generator data category and modeled as MCTYPE=4 in IEC data file.)

Each data record has the following format:

I, ID, ILR2IR_1, R2X_1, ILR2IR_2, R2X_2, ILR2IR_0, R2X_0

where:

I	Machine bus number
ID	Machine ID
ILR2IR_1	Ratio of Positive Sequence locked rotor current to rated current, no default allowed.
R2X_1	Ratio of Positive Sequence resistance to reactance =0.1 by default
ILR2IR_2	Ratio of Negative Sequence locked rotor current to rated current ILR2IR_2= ILR2IR_1 by default
R2X_2	Ratio of Negative Sequence resistance to reactance R2X_2= R2X_1 by default
ILR2IR_0	Ratio of Zero Sequence locked rotor current to rated current ILR2IR_0= ILR2IR_1 by default
R2X_0	Ratio of Zero Sequence resistance to reactance R2X_0= R2X_1 by default

Wind Power Station Doubly Fed Asynchronous Generator Data

(These equipment are specified as part of Generator data category, and modeled as MCTYPE=5 in IEC data file.)

Each data record has the following format:

I, ID, IMAX_1, R2X_1, IMAX_2, R2X_2, IMAX_0, R2X_0, KWD, IKWDmax, IKWDmin

where:

I	Machine bus number
ID	Machine ID
IMAX2IR_1	Ratio of Positive Sequence highest instantaneous short circuit value in case of a three-phase short circuit to rated current, no default allowed

R2X_1	Ratio of Positive Sequence resistance to reactance. R2X_1=0.1 by default
IMAX2IR_2	Ratio of Negative Sequence highest instantaneous short circuit value in case of an unbalance short circuit to rated current, IMAX2IR_2= IMAX2IR_1 by default
R2X_2	Ratio of Negative Sequence resistance to reactance. R2X_2= R2X_1 by default
IMAX2IR_0	Ratio of Zero Sequence highest instantaneous short circuit value in case of an unbalance short circuit to rated current, IMAX2IR_0= IMAX2IR_1 by default
R2X_0	Ratio of Zero Sequence resistance to reactance. R2X_0= R2X_1 by default
KWD	Factor for the calculation of the peak short-circuit current, KWD=1.7 by default
IkWDmax	Ratio of maximum steady state short circuit current to rated current, IkWDmax = IMAX2IR_1 by default. This is used in calculation of symmetrical breaking current.
IkWDmin	Ratio of minimum steady state short circuit current to rated current, IkWDmin = IMAX2IR_1 by default. This is used in calculation of symmetrical breaking current.

Wind and Photovoltaic Power Station Generator with Full Size Converter Data

(These equipment are specified as part of Generator data category, and modeled as MCTYPE=6, 7 in IEC data file.)

Each data record has the following format:

I, ID, IMAX_1, R2X_1, IMAX_2, R2X_2, IMAX_0, R2X_0, IkPFmax, IkPFmin

where:

I	Machine bus number
ID	Machine ID
IMAX2IR_1	Ratio of Positive Sequence short circuit current value to rated current, no default allowed
R2X_1	Ratio of Positive Sequence resistance to reactance. R2X_1=0.1 by default

IMAX2IR_2	Ratio of Negative Sequence short circuit current value to rated current, IMAX2IR_2= IMAX2IR_1 by default
R2X_2	Ratio of Negative Sequence resistance to reactance. R2X_2= R2X_1 by default
IMAX2IR_0	Ratio of Zero Sequence short circuit value in case of an unbalance short circuit to rated current, IMAX2IR_0= IMAX2IR_1 by default
R2X_0	Ratio of Zero Sequence resistance to reactance. R2X_0= R2X_1 by default
IkPFmax	Ratio of maximum steady state short circuit current to rated current, IkPFmax = IMAX2IR_1 by default. This is used in calculation of symmetrical breaking current.
IkPFmin	Ratio of minimum steady state short circuit current to rated current, IkPFmin = IMAX2IR_1 by default. This is used in calculation of symmetrical breaking current.

Data records for generating units and motors may be entered in any order and using a free format with blanks or commas separating each data item in each record. These records are terminated with a record specifying machine bus number I value of zero. Following is an example of IEC data for IEC 60909-4, Section 6 network. (Refer to the PAG for details.)

```
1  Q1 2
5  Q2 2
6  G3 1 10.5 0.0 0.8
41 G1 1 21.0 0.0 0.85 0 1 4 41 0 T1 12
31 G2 1 10.5 7.5 0.9  0 2 3 31 0 T2
0 / END OF GSU, EQV, GEN, AND INDUCTION MACHINE DATA
0 / END OF TRANSFORMER DATA
7  M1 1 0.88 97.5
7  M2 2 0.89 96.8
7  M3 2 0.89 96.8
0 / END OF INDUCTION MACHINE DATA
Q
```

or

```
IECFILEVRSN=2
1  Q1 2
5  Q2 2
6  G3 6 10.5 0.0 0.8
41 G1 4 21.0 0.0 0.85 0 1 4 41 0 T1 12
31 G2 5 10.5 7.5 0.9  0 2 3 31 0 T2
0 / END OF GSU, EQV, GEN, AND MOTOR DATA
0 / END OF TRANSFORMER DATA
7  M1 1 0.88 97.5
```

7 M2 2 0.89 96.8
7 M3 2 0.89 96.8
0 / END OF INDUCTION MACHINE DATA
41 G1 5.0 0.2
0 / END OF WIND POWER STATION ASYNCHRONOUS GENERATOR DATA
31 G2 6.0 0.3
0 / END OF WIND POWER STATION DOUBLY FED ASYNCHRONOUS GENERATOR DATA
6 G3 7.0 0.4
0 / END OF WIND AND PHOTOVOLTAIC POWER STATION GENERATOR WITH FULL SIZE CONVERTER DATA
Q